

COM668 Project

System design and technology selection

System Design and technology selection

- Informal architecture
- Client-server model
- Unified Modelling Language
- Data design
- Design patterns
- Technology selection

System structure

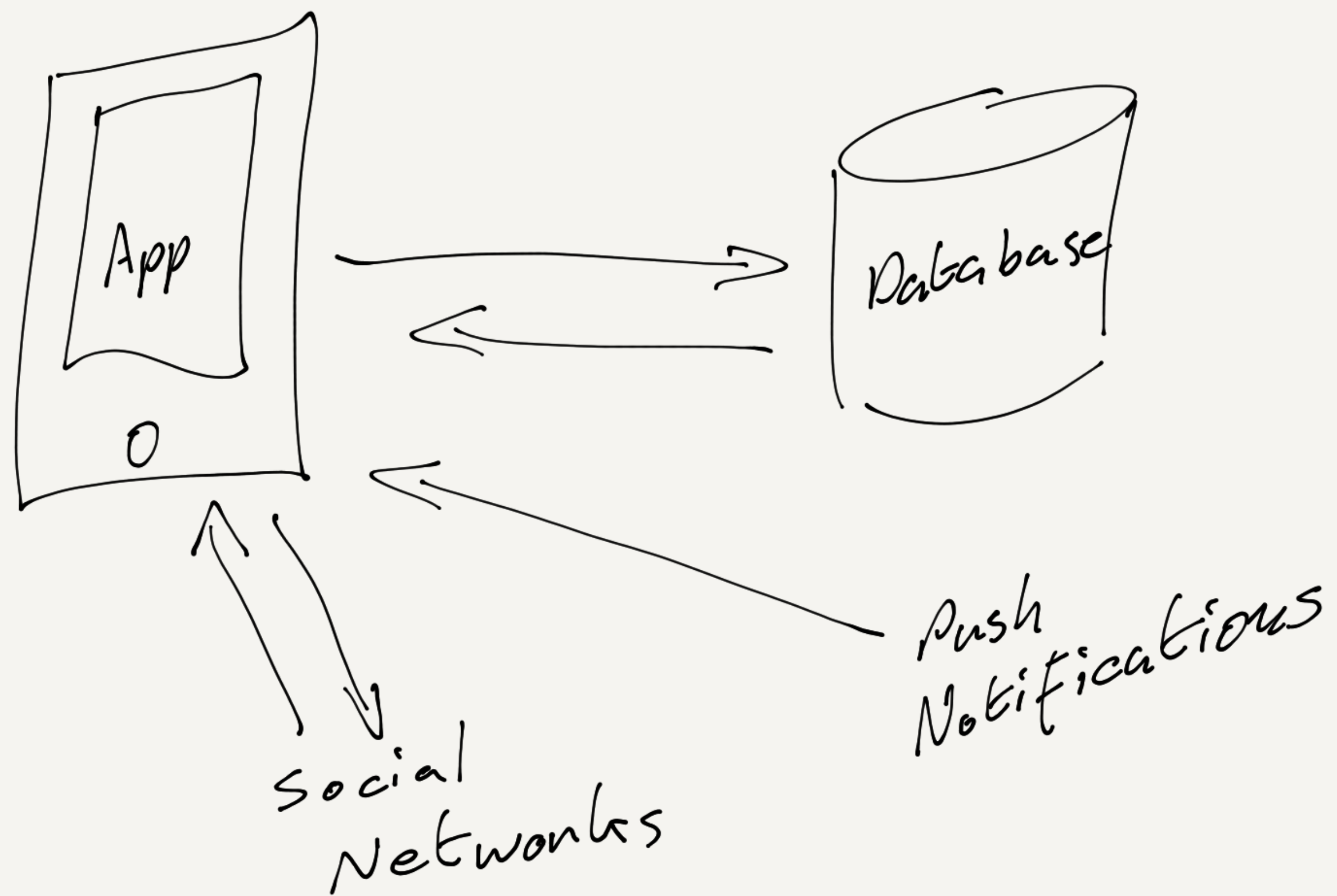
- Understand the structure of your system
- Identify the elements that will form that structure
- Model the relationships between the elements
- Architect a complete and robust system
- Select the technologies needed to realise your solution

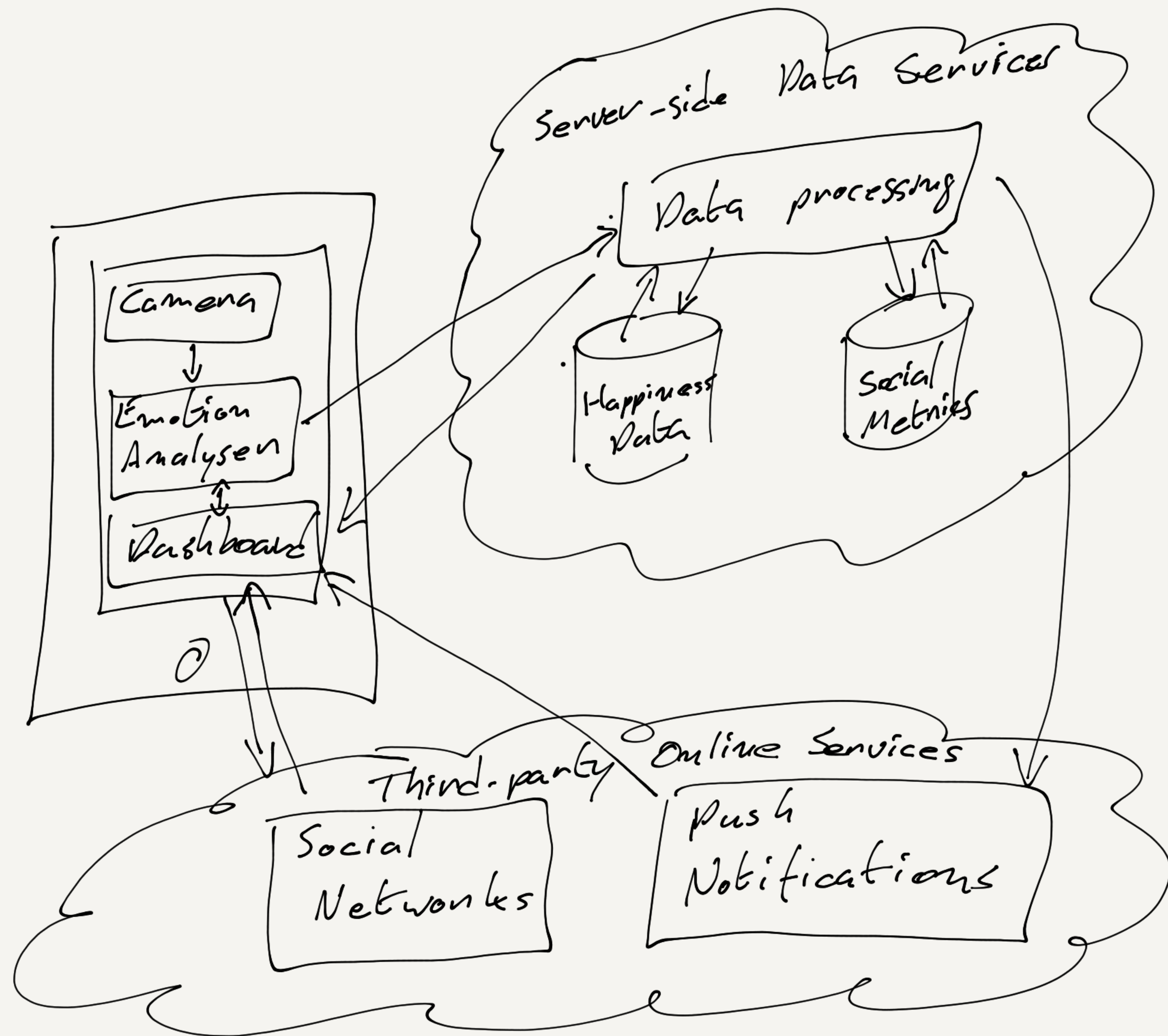
Informal Architecture

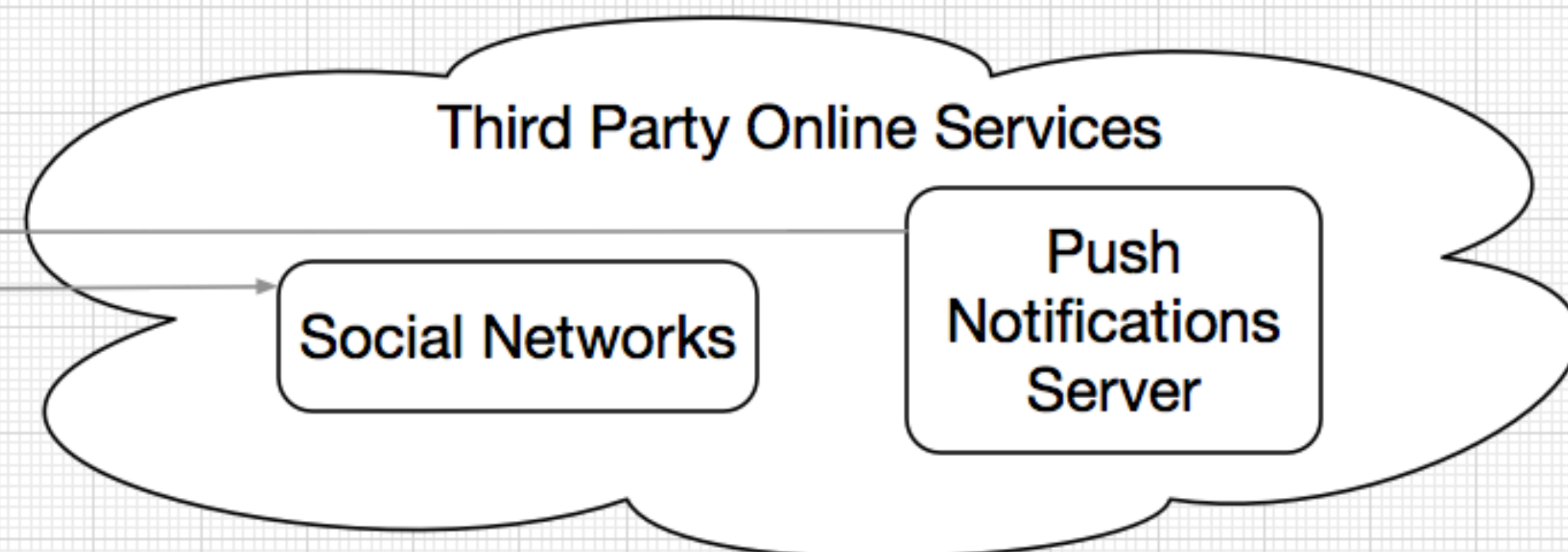
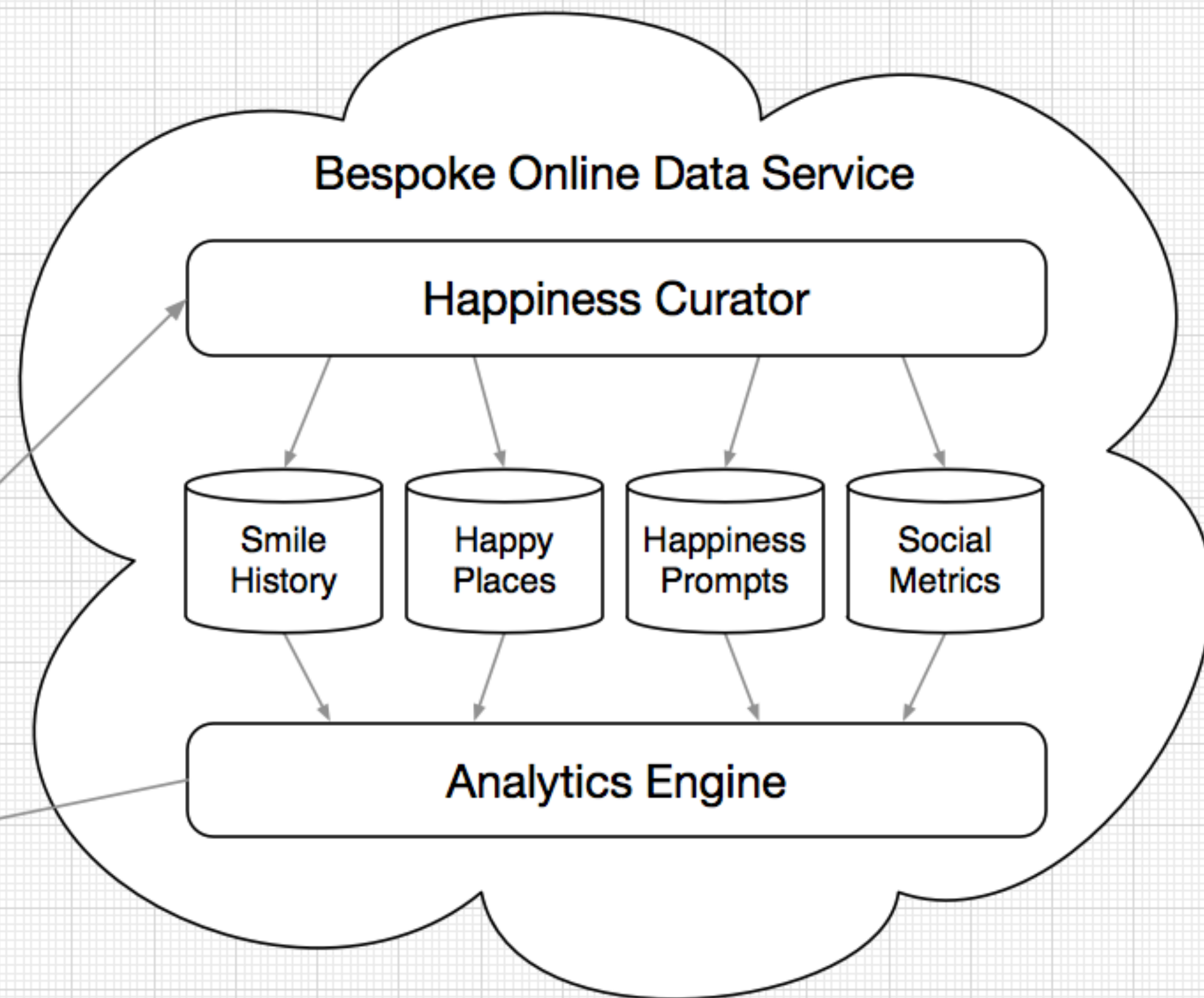
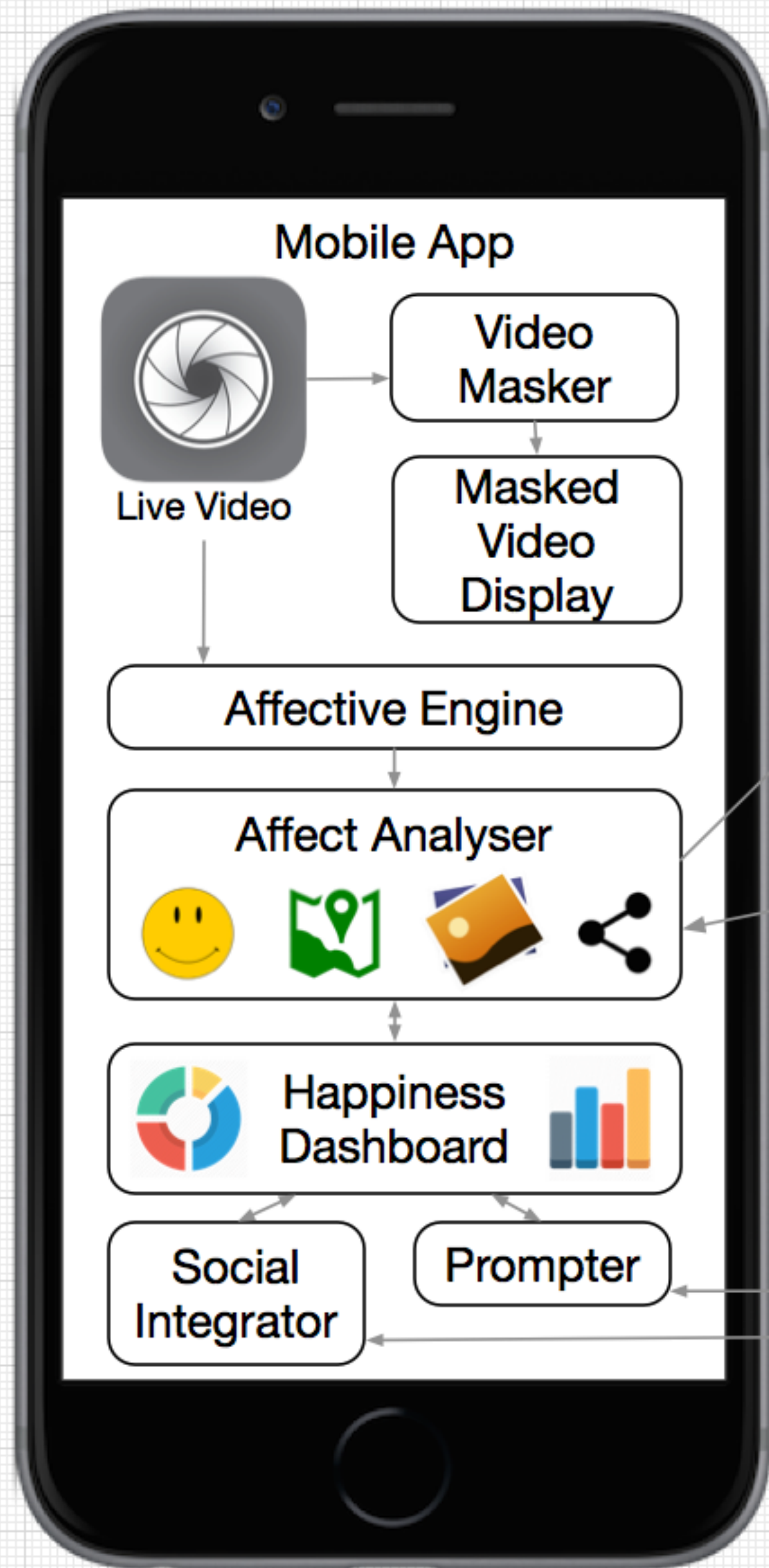
Quick and human

From sketches to diagrams

- Can be useful to start a design with informal sketches
- Sketches are quick, creative, meaningful and disposable
- Can be refined as diagrams later for presentation purposes
- Explored with standardise diagrams to convey understanding







Client-server model

For The Website

Client-server model

- One of a number of perspectives that can be taken
- An established and straightforward model
- Easy to understand
- Useful as a starting point for identifying technologies

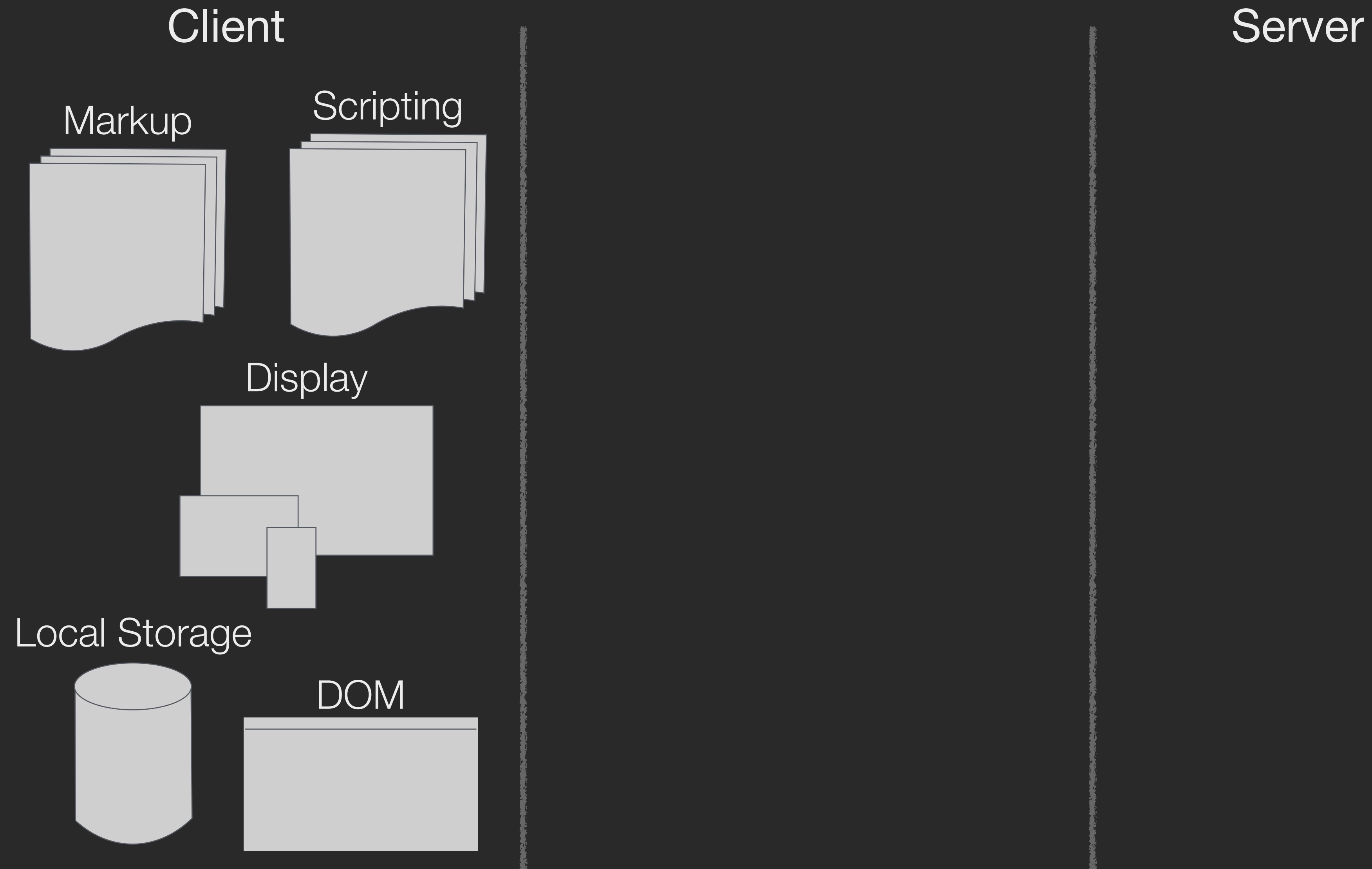
Client-server model

Client

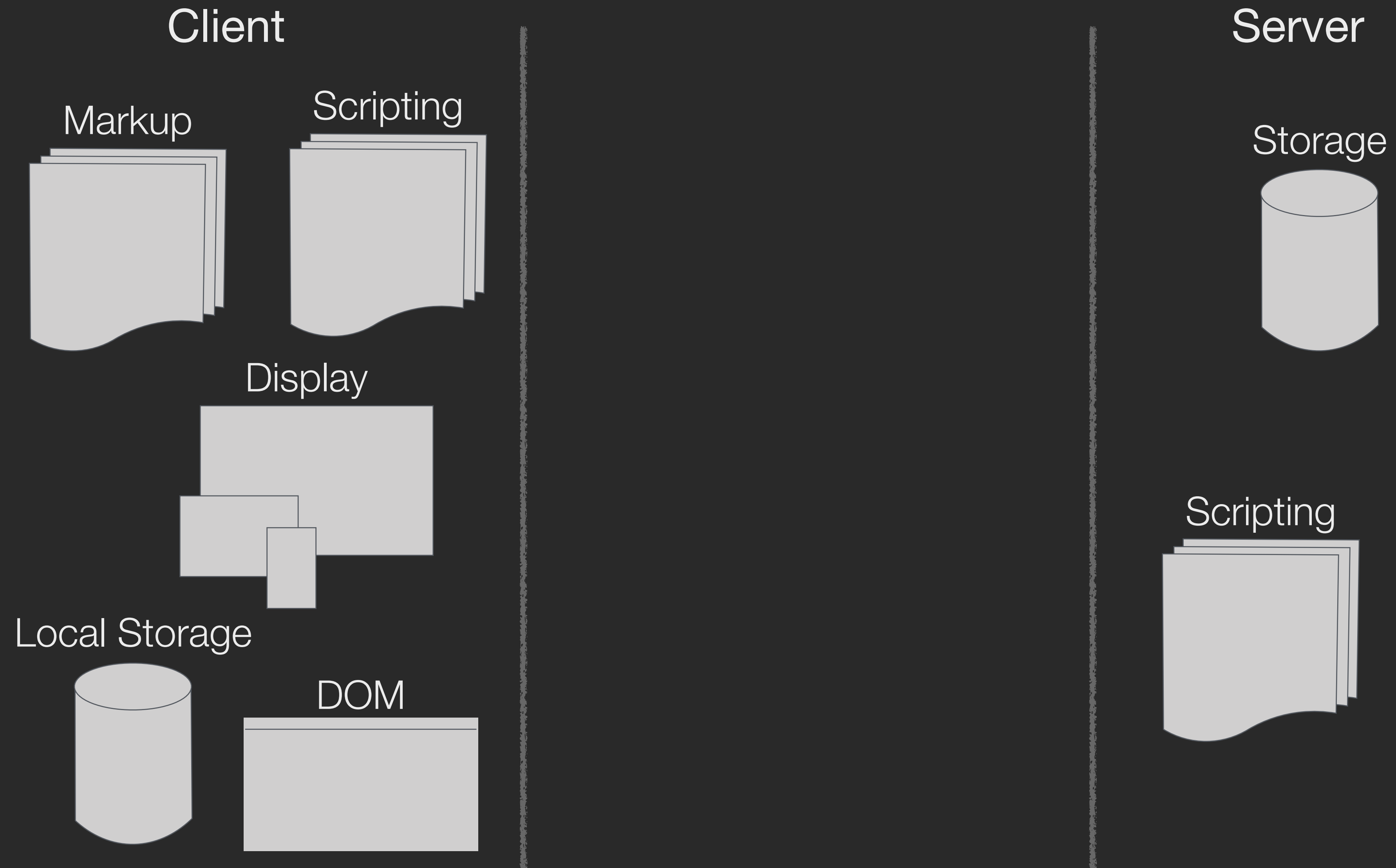
Server



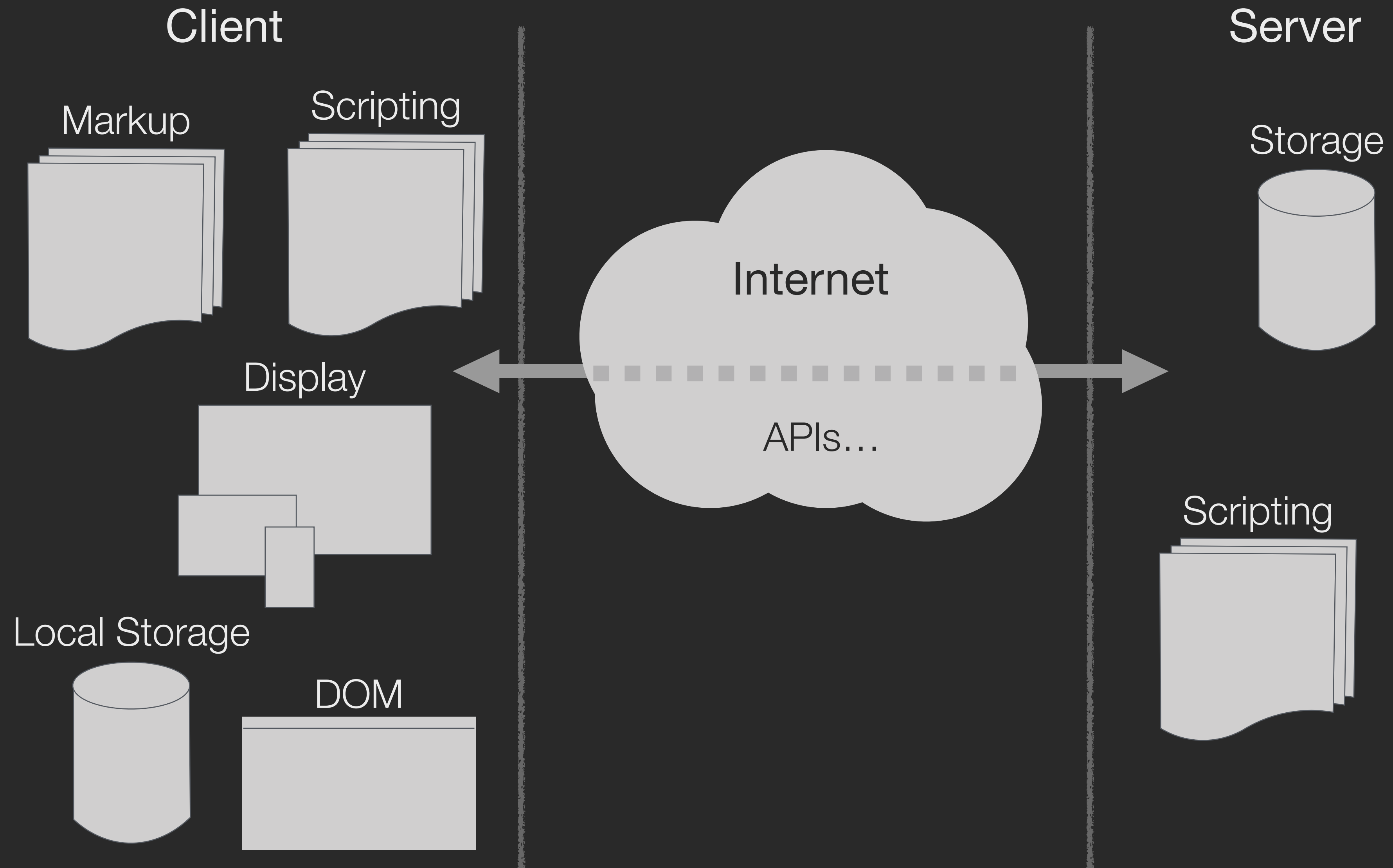
Client-server model



Client-server model



Client-server model



Unified Modelling Language

Diagrams for everything

UML

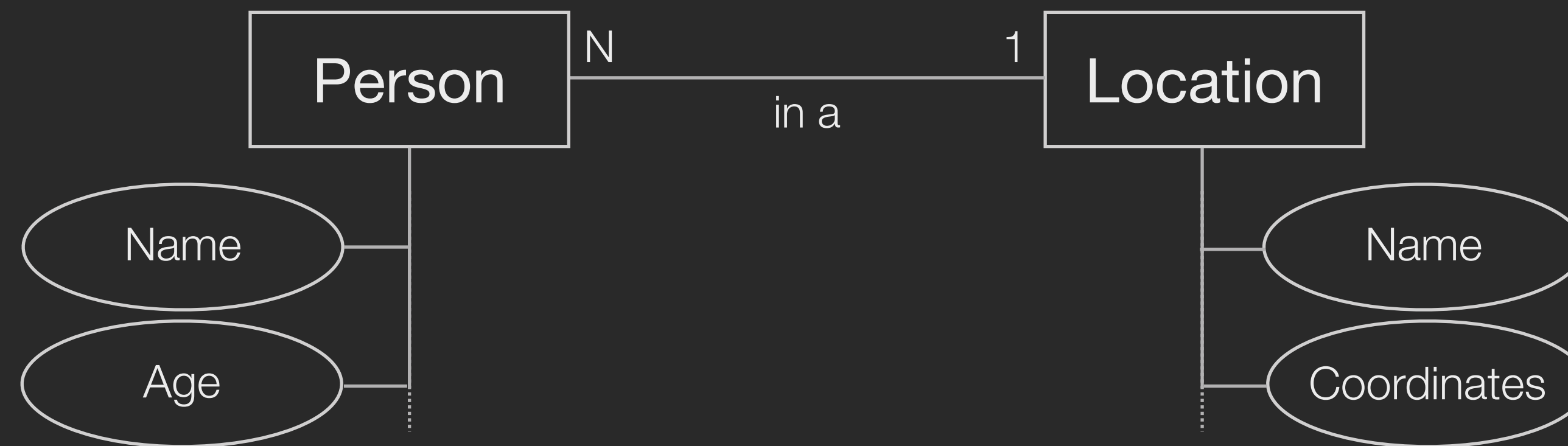
- General purpose language for modelling object-oriented systems
- Used to design and visualise a system from various perspectives
- Wide range of diagram types to help model these perspectives
 - Structure:
Class Diagram, Object Diagram, Component Diagram, Composite Structure Diagram, Package Diagram and Deployment Diagram
 - Behaviour:
Use Case Diagram, Activity Diagram and State Machine Diagram
 - Interaction:
Sequence Diagram, Communication Diagram, Timing Diagram and Interaction Overview Diagram
- Some diagram types helpful even for a non object-oriented approach

Data design

Getting data structures correct early is critical

Data design

- Can be critical to have a stable data design before coding
- Database Entity Relationship modelling
 - Intuitive data diagramming technique
 - Helps to identify database tables, fields and relationships
 - Database design should be normalised



- Also consider the design of other data structure you plan to use

Design patterns

Reusable building blocks

Design patterns

- Low-level, reusable and established designs
- Common design patterns
 - MVC, Grid patterns, Navigation menu patterns, Lists patterns, Headers and footers patterns, Form patterns...
- Facilitate reuse
 - Saves design and development time
 - Reduces potential for errors
- Promotes consistent design
- Identify design patterns that you can use

Technology selection

Select and justify the technologies you will use

Technology selection

- Consider each element of the system
- Identify candidate technologies that could be used
 - Programming languages
 - Database management systems
 - Markup and scripting languages
 - Frameworks, libraries, APIs and services
- Do not provide detailed descriptions of all candidate technologies
- Compare and evaluate their suitability for your project
- Specify the technologies you will use
- Justify your decisions

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